Git & GitHub Complete Notes (Comprehensive)

All commands with 1-2 line explanations. Use as a printable quick reference.

1. Basic Setup

Command	Purpose
git --version	Check installed Git version.
git config --global user.name "Your Name"	Set global username for all commits.
git config --global user.email " “	Set global email for commits.
git config --global color.ui auto	Enable colored Git output.
git config --list	Show all Git configuration values.

2. Repository Initialization

Command	Purpose
git init	Initialize a new Git repository in current folder.
git clone <URL>	Clone a remote repository to local machine.
git clone <URL> <foldername>	Clone repository into specified folder.
git fork (GitHub UI)	Create your copy of a repository on GitHub (via website).

3. File Tracking & Staging

Command	Purpose
git status	Show working tree status (modified/untracked files).
git add <file>	Stage a specific file for commit.
git add .	Stage all changes in current directory.
git add -A	Stage all changes (including deletions).
git rm <file>	Remove file from index and working tree (delete).
git rm --cached <file>	Remove file from index but keep it locally.
git mv <old> <new>	Rename/move a file and stage the change.

4. Commit Commands

Command	Purpose
git commit -m "message"	Create a commit from staged changes with a message.
git commit -am "message"	Stage tracked files and commit in one command.
git commit --amend	Modify the last commit message or include additional changes.
git log	Show commit history for current branch.
git log --oneline --graph --decorate --all	Compact, graphical commit history view.
git show <commit>	Show details (diff, metadata) of a specific commit.

5.Branching

Command	Purpose
git branch	List local branches.
git branch <name>	Create a new branch.
git checkout <branch>	Switch to an existing branch.
git checkout -b <branch>	Create a new branch and switch to it.
git branch -d <branch>	Delete a branch that is merged.
git branch -D <branch>	Force-delete a branch (even if unmerged).

6. Merging & Rebasing

Command	Purpose
git merge <branch>	Merge specified branch into current branch.
git merge --abort	Abort an in-progress merge and return to previous state.
git rebase <branch>	Reapply commits on top of another base branch.
git rebase --continue	Continue rebasing after resolving conflicts.
git rebase --abort	Abort rebase and return to original branch state.

7. Undoing / Fixing Mistakes

Command	Purpose
git restore <file>	Restore working directory file to last committed state.
git restore --staged <file>	Unstage a file (remove from staging area).
git reset HEAD <file>	Unstage file (alternative to restore --staged).
git reset --soft HEAD~1	Undo last commit but keep changes staged.
git reset --mixed HEAD~1	Undo last commit, keep changes in working dir (default).
git reset --hard HEAD~1	Discard last commit and all changes (dangerous).
git revert <commit>	Create a new commit that reverses changes from given commit.

8. Diff Commands

Command	Purpose
git diff	Show unstaged changes between working dir and index.
git diff --staged	Show differences between staged changes and last commit.
git diff <branch1> <branch2>	Compare two branches' differences.
git diff <commit1> <commit2>	Compare two commits.

Command	Purpose
git remote -v	List configured remote URLs.
git remote add origin <URL>	Add a remote repository (e.g., GitHub).
git remote remove origin	Remove remote connection.
git remote set-url origin <URL>	Change the remote URL for origin.

10. Push & Pull

Command	Purpose
git push -u origin main	Push local branch to remote and set upstream.
git push	Push committed changes to the configured remote.
git push origin <branch>	Push specified branch to remote.
git pull	Fetch from remote and merge into current branch.
git fetch	Download objects/refs from remote without merging.
git fetch –all	Fetch all remotes.
git pull origin main	Fetch and merge changes from main on origin.

Command	Purpose
git stash	Save uncommitted changes to a stack and clean working dir.
git stash list	List stashed entries.
git stash apply	Apply last stash to working dir (keep stash).
git stash pop	Apply last stash and remove it from stash list.
git stash drop	Remove a specific stash.
git stash clear	Remove all stashes.

Command	Purpose
git tag	List tags in repository.
git tag <name>	Create a lightweight tag.
git tag -a <name> -m "message"	Create an annotated tag with message.
git push origin <tag>	Push a specific tag to remote.
git push --tags	Push all local tags to remote.

13. GitHub CLI (gh)

Command	Purpose
gh auth login	Authenticate GitHub CLI with your account.
gh repo create	Create a new repository via CLI.
gh repo clone <repo>	Clone using GitHub CLI.
gh issue create	Create an issue from the terminal.
gh pr create	Create a pull request via CLI.
gh pr merge	Merge a pull request via CLI.

7.

14. Pull Request Workflow

Command	Purpose
git checkout -b feature/<name>	Create and switch to a feature branch.
git push origin feature/<name>	Push feature branch to remote.
Create PR on GitHub	Open a pull request for review & merge (via web UI).
git pull origin main	Update local main after PR merge.

Command	Purpose
git clean -n	Show which untracked files would be removed (dry run).
git clean -f	Delete untracked files.
git clean -fd	Delete untracked files and directories.
git clean -fx	Delete untracked files including ignored files.

Command	Purpose
.gitignore	File listing patterns for files/folders Git should ignore.
git rm --cached <file>	Remove a previously committed file while keeping local copy.

Command	Purpose
Fork (GitHub UI)	Create your copy of upstream repository on GitHub.
git clone <forked_url>	Clone your fork locally.
git remote add upstream <original_url>	Add upstream remote to track original repo.
git fetch upstream	Fetch changes from upstream repository.
git merge upstream/main	Merge upstream changes into your local main.

Command	Purpose
ssh-keygen -t ed25519 -C "email@examp.co	m"Generate a new SSH keypair.
cat ~/.ssh/id_ed25519.pub	Display the public key to add to GitHub.
ssh-add ~/.ssh/id_ed25519	Add private key to SSH agent.
git clone git@github.com:user/repo.git	Clone using SSH protocol.

Command	Purpose
git reflog	Show history of HEAD references (recover lost commits).
git bisect	Binary search to find commit that introduced a bug.
git shortlog	Summarize git log by author.
git archive --format=tar --output=repo.tar HEAD	Create an archive of current HEAD.